

ONLINE CLOCK AND CALENDAR WEB

APP AI23431- WEB TECHNOLOGY AND

MOBILE APPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

JESHWANTH P

(241801106)

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE
AND DATA SCIENCE**

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(AUTONOMOUS),

CHENNAI-602 105

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RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled “**ONLINE CLOCK AND CALENDAR WEB APP**” is the Bonafide work of JESHWANTH P (241801106) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Dr. J.M. Gnanasekar M.E., Ph.D.,
Professor and Head
Department of AI&DS
Rajalakshmi Engineering College
Chennai – 602 105

SIGNATURE

Mr. G. Thiyagarajan
Assistant Professor
Department of AI&DS
Rajalakshmi Engineering College
Chennai – 602 105

Submitted for the project viva-voce examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

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JESHWANTH P(241801106)

DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with an essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and time management: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Data Science Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open- ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and ML professional.

COURSE OBJECTIVE

- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOME

On completion of course students will be able to,

- CO 2: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.
- CO 3: Design and implement dynamic web page with validation using javascript objects and by applying different event handling mechanism.
- CO 4: Design and implement simple webpage to learn JSP and Servlet.
- CO 5: Design and implement simple Application Design.
- CO 6: Identify various concepts of mobile programming that make it unique from programming for other

CO-PO-PSO Mapping

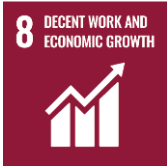
CO	PO 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	P O 10	P O 11	P O 12	PS O 1	PS O 2	PS O 3
CO 1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO 2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO 3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO 4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO 5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3:

Substantial (High) No correlation: “-”

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
 SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
 SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
 SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
 SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

This project is an Online Clock and Calendar Web App designed to help users view real-time date and time information, manage events, and organize schedules efficiently through a user-friendly web interface. The application provides features such as digital and analog clock displays, interactive calendars, event scheduling, reminders, and timezone support. Users can log in securely, create and manage events, set reminders, and access their personalized dashboard. The system is developed using Java, JSP, JDBC, and MySQL with an MVC architecture to ensure maintainability and efficient data management. Overall, the project demonstrates how modern web technologies can be used to develop an effective online clock and calendar management platform.

This project is an Online Clock and Calendar Web App web application designed to simplify the process of recording, managing, and analyzing personal or organizational date, time, and event information. It provides a convenient platform where users can log their event and events, categorize them, and monitor their time and scheduling health through an intuitive dashboard.

The application allows users to create accounts, log in, and access personalized clock and calendar dashboards. Users can add event and event entries with categories and dates, view summaries of their spending patterns, and filter records by date range or category for quick analysis.

The system is developed using Java, JSP, and MySQL, and follows a structured MVC approach to keep different parts of the application organized. Basic security features such as user authentication and safe database operations are implemented to ensure reliability and data integrity.

Overall, the project demonstrates how web technologies can be used to build a practical and user-friendly platform for tracking, categorizing, and reporting personal and organizational time and scheduling data.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JSP	Java Server Pages
JDBC	Java Database Connectivity
SQL	Structured Query Language
DBMS	Database Management System
UI	User Interface
UX	User Experience
HTTP	HyperText Transfer Protocol

CHAPTER 1

INTRODUCTION

1.1 General

In today's digital era, managing personal and organizational time managements has become increasingly important. People often struggle to keep track of their daily events, savings, and event due to the lack of a structured system. An online Online Clock and Calendar Web App provides users with quick access to their time and scheduling data, making reminding, planning, and spending analysis more efficient and convenient.

This project focuses on the development of a web-based Online Clock and Calendar Web App that allows users to record date, time, and event information, categorize events, view graphical summaries, and manage their reminders. The system is designed to be user-friendly and accessible, enabling users to perform various actions such as adding event and event entries, filtering records by date or category, and generating time and scheduling reports.

1.2 Need for Online Clock and Calendar Web App System

Traditional methods of event management, such as maintaining manual ledgers or using spreadsheets, are often error-prone, time-consuming, and difficult to analyze. Individuals and small businesses need an accurate and up-to-date record of their date, time, and event information to make informed decisions.

An Online Clock and Calendar Web App system addresses these challenges by providing a centralized digital platform where users can log, categorize, and analyze their date, time, and event information. It reduces dependency on manual record-keeping, improves accessibility, and allows users to compare event versus events easily. Additionally, it ensures that records are always up to date, making time and scheduling management more reliable and efficient.

1.3 Scope of the System

The system is designed to serve as a comprehensive platform for personal and organizational event management. It allows users to register, log in, and manage their personalized time and scheduling dashboards. Users can add calendar events and reminders with categories, dates, and descriptions, and view summaries and charts of their time and scheduling activity. The system also provides monthly and category-wise reports to help users plan their reminders.

Furthermore, users can set monthly reminder limits and receive alerts when spending exceeds the configured threshold. The system is intended for individual users and small businesses and can be extended in the future with additional features such as automated bank statement import, multi-currency support, and mobile application integration.

1.4 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the online clock and calendar web app system.

Chapter 1: provides the introduction to the project, including the general overview, need for the system, and scope of the project.

Chapter 2: presents the literature survey, which discusses existing clock and calendar management platforms and related work.

Chapter 3: explains the problem formulation and objectives of the proposed system.

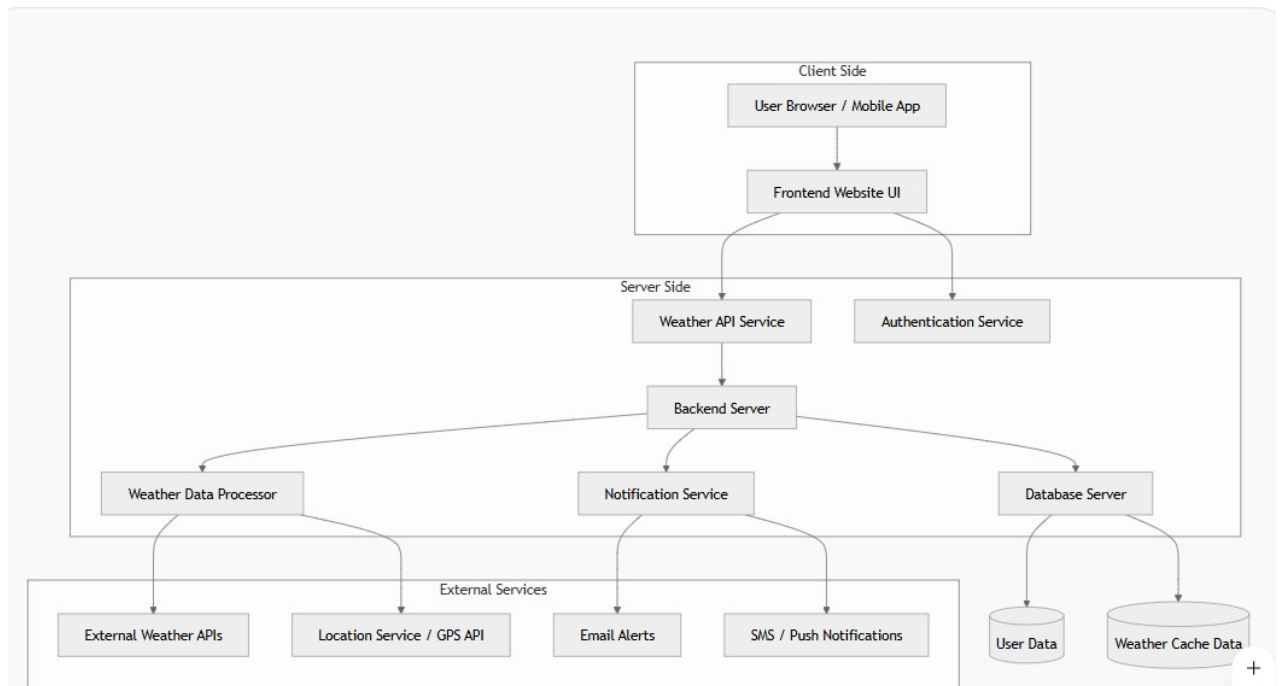
Chapter 4: describes the system design, including system architecture, database design, and system requirements.

Chapter 5: discusses the implementation of the system and explains the different modules.

Chapter 6: presents the results and discussion, including system outputs and performance evaluation.

Chapter 7: concludes the project and suggests possible future enhancements.

System architecture diagram:s



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The literature survey explores existing systems and approaches used in personal time management and clock and calendar management platforms. With the rapid growth of internet usage, digital solutions have become the preferred method for managing time and scheduling data. This section examines how current platforms function, the technologies they use, and the challenges they face. Understanding these aspects helps in designing a system that is both efficient and user-friendly while addressing the limitations of existing solutions.

2.2 Existing Online Clock and Calendar Web App Systems

Modern clock and calendar management platforms provide users with the ability to record, view, and analyze their date, time, and event information conveniently. These systems typically offer features such as category-based filtering, monthly summaries, event reminders and alerts, and graphical expenditure displays. Users can also set spending limits and receive notifications when thresholds are crossed.

While these platforms are powerful and widely used, they often come with certain limitations. Many systems are designed for large-scale commercial use, making them complex and sometimes difficult for new users to navigate. In addition, heavy reliance on subscription models can increase operational costs. Some platforms also lack customization, making it difficult for users to tailor categories or reporting formats to their personal needs, which can affect the overall user experience.

2.3 Related Work

Previous studies and projects in the domain of web-based online clock and calendar web app systems have focused on improving usability, efficiency, and data accuracy. Researchers have explored the

systems. Database-driven applications are commonly used to store and manage large volumes of time and scheduling event data, ensuring that users can access updated and accurate records.

Security and user authentication are also key areas of focus in related work, with systems implementing login mechanisms and session management to protect user data. Some solutions incorporate advanced features such as AI-based event prediction and automated bank statement parsing to enhance user experience. However, these features often increase system complexity and may not be necessary for all users, especially in smaller-scale applications.

2.4 Summary

The analysis of existing platforms and related work highlights the importance of providing a balanced system that combines functionality with simplicity. While many advanced systems offer a wide range of time and scheduling analytics features, there is still a need for applications that are easy to use, efficient, and focused on essential event management functionalities.

Based on this study, the proposed system aims to deliver a streamlined clock and calendar management platform that enables users to add, search, and manage time and scheduling records with ease. The focus is on creating a reliable and user-friendly application that meets core time and scheduling management requirements without unnecessary complexity.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

In the current digital environment, managing personal and organizational time managements effectively is often challenging. Users may struggle to track daily spending, monitor savings goals, and analyze category-wise expenditure, while small businesses face significant challenges due to a lack of structured time and scheduling record-keeping systems.

Existing clock and calendar management platforms, although feature-rich, are often complex and may include unnecessary functionalities for basic users. They can also rely heavily on subscription-based services, which may increase operational costs for smaller applications. Additionally, many systems do not provide a simple and centralized interface for managing custom categories and personalized event reminders and alerts.

Therefore, there is a need for a streamlined and user-friendly web-based system that allows users to easily add, view, and manage date, time, and event information across multiple categories, while providing timely summaries and event reminders and alerts in an accessible format.

3.2 Objectives of the System

The main objective of the system is to develop a web-based application that simplifies the process of recording and managing date, time, and event information. The specific objectives are as follows:

- To design and develop a user-friendly interface for adding and viewing calendar events and reminders
- To enable users to search and filter time and scheduling records based on category, date range, or event type
- To provide functionality for users to save and manage their favourite locations
- To allow users to view detailed time and scheduling summaries including total event, total events, balance, and category-wise breakdown
- To implement a reminder alert feature to notify users when their spending exceeds the configured monthly limit
- To develop a secure login and registration system using session management
- To create a structured and maintainable system using appropriate design patterns and API

integration techniques

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design is an important phase in software development where the overall structure and components of the system are planned. It defines how different modules interact with each other and how data flows within the system. A well-designed system improves efficiency, reliability, and ease of maintenance.

The Online Clock and Calendar Web App system is designed as a web-based application that allows users to add, search, and manage date, time, and event information. It provides features such as user registration, login, event entry management, category filtering, and report generation. The system follows a client-server MVC architecture where users interact through web pages, while the backend processes requests and communicates with the database.

4.2 System Architecture

The system follows a three-layer architecture consisting of the presentation layer, application layer, and data layer.

Presentation Layer

This layer represents the user interface where users interact with the system. It includes web pages such as the homepage, login page, registration page, clock and calendar dashboard page, and event history page. These pages are developed using JSP, HTML, and CSS, and are responsible for displaying time and scheduling data to users and collecting event input.

Application Layer

This layer handles the core logic of the system. It processes user requests, manages sessions, performs input validation, calculates event summaries and event reminders and alerts, and controls the flow of data between the presentation layer and the database.

The backend is implemented using Java Servlets, where each servlet is responsible for a specific functionality:

- PropertyServlet → Displays all event records
- SearchServlet → Handles property search
- AddEventServlet → Adds new event or event entries
- PropertyDetailServlet → Shows detailed property information
- InquiryServlet → Manages user inquiries
- LoginServlet & RegisterServlet → Handle authentication
- DeleteEventServlet → Deletes event or event entries

This layer ensures proper execution of all operations and maintains system functionality.

Data Layer

This layer is responsible for storing and managing all application data. The system uses a MySQL database connected through JDBC for data operations, ensuring accurate and persistent storage of all date, time, and event information and user information.

The database consists of three main tables:

- **users** → Stores user details such as name, email, and password
- **events** → Stores property information including title, price, location, description, image, and user ID
- **inquiries** → Stores reminder alert thresholds configured by each user

4.3 Database Design

The database is designed to efficiently store and manage data related to users, date, time, and event information, and reminder configurations. Each table is structured with appropriate fields and relationships to ensure data integrity and efficient querying.

- The **users** table stores registered user information
- The **events** table stores all date, time, and event information and is linked to users through a foreign key
- The **inquiries** table stores reminder alert configurations and thresholds set by users

4.4 System Requirements

System requirements define the hardware and software resources needed to run the application successfully.

4.4.1 Hardware Requirements

The system does not require high-end hardware and can run on standard computers.

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB
- **Storage:** 500 GB Hard Disk or SSD
- **Input Devices:** Keyboard and Mouse
- **Internet Connection:** Required

4.2.1 Software Requirements

The system is developed using commonly available tools and technologies.

- 5 **Operating System:** Windows 10 or later
- 6 **Frontend Technologies:** HTML, CSS, JSP
- 7 **Backend Technology:** Java Servlets
- 8 **Database:** MySQL
- 9 **Database Connectivity:** JDBC
- 10 **Web Server:** Apache Tomcat 9.0
- 11 **Development Environment:** Eclipse IDE for Enterprise Java
- 12 **Web Browser:** Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The system is developed using a structured approach based on the Model-View-Controller (MVC) architecture. This methodology helps in separating the user interface, business logic, and data handling, making the system easier to develop and maintain.

- **Model:** Represents the data and business logic using Java classes such as [Property](#), [User](#), and [Inquiry](#).
- **View:** Consists of JSP pages that display data to users and collect input.
- **Controller:** Implemented using Java Servlets, which handle user requests, process data, and control the flow between the view and model.

The development process involves designing the database schema, creating model classes for events and users, implementing servlets for all CRUD functionalities, and developing JSP pages for user interaction. JDBC is used to connect the application with the MySQL database.

5.2 Module Descriptions

The system is divided into several modules, each responsible for a specific functionality.

5.2.1 User Module

This module handles user registration, login, and session management.

- New users can register by providing details such as name, email, and password
- Existing users can log in using valid credentials
- Session management is implemented using [HttpSession](#)
- User data is stored securely in the database

5.2.2 Event Management Module

This module allows users to add, view, and delete date, time, and event information (both event and event entries).

- Users can save new locations by entering city name or allowing GPS-based detection
- Saved locations are stored in the database for quick access
- Users can delete their saved locations
- All events with totals and category breakdown are displayed on the dashboard

5.2.3 Event Search Module

This module provides functionality to search and filter event records based on category, date range, or event type.

- Users can enter a city name or postal code
- The system performs a search using SQL **LIKE** queries
- Matching event records are displayed dynamically in a tabular format
- Helps users quickly find specific events or analyse spending in a particular period

5.2.4 Event Details Module

This module displays detailed meteorological information for a selected location.

- Shows complete details of a selected event including amount, category, date, description, and running balance
- Allows users to review event history and detailed summaries before planning their reminder
- Improves user experience by providing detailed and accurate time and scheduling records

5.2.5 Reminder Alert Module

This module enables users to set and manage event reminders and alerts.

- Users can configure monthly reminder limits for specific categories such as food, transport, or entertainment
- Alert preferences are stored in the database
- Helps users stay informed about overspending and take corrective time and scheduling actions

5.2.6 Login and Authentication Module

This module ensures secure access to the system.

- Validates user credentials during login
- Uses session management to maintain user state
- Prevents unauthorized access to restricted features
- Handles login and logout operations

5.2.7 Delete Event Module

This module allows users to delete their recorded events from the system.

- Only the account owner can delete their saved locations
- Related reminder alert configurations are also updated when a event is removed
- Maintains data consistency and integrity

CHAPTER 6

RESULTS AND DISCUSSION

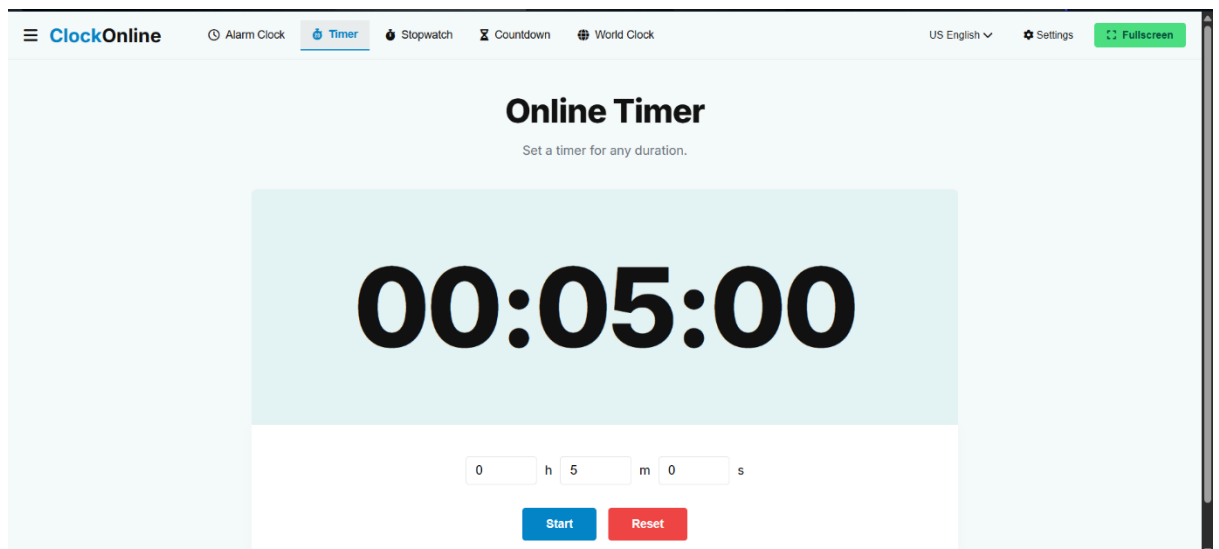
6.1 System Output Screens

The developed system was tested by executing all major functionalities, and the outputs were verified through different user interactions. The application successfully displays time and scheduling summaries, event records, and reminder alert notifications, and responds correctly to all user inputs.

The main output screens of the system include:

- **Homepage (Event Dashboard):**

Displays the clock and calendar dashboard showing total event, total events, current balance, and a category-wise event breakdown. Data is fetched from the database and shown in a structured card and chart format.



- **Login Page:**

Allows registered users to log in using their email and password. Proper validation messages are displayed for incorrect credentials.

This also includes:

- **Transaction Details Page:**

Shows complete meteorological data for a selected city, including temperature trend graphs, UV index, visibility, and sunrise/sunset times.

- **Add Transaction Page:**

Allows users to add a new event or event event with category, amount, date, and description.

- **Reminder Alerts Page:**

Enables users to set a monthly reminder limit. A confirmation message is displayed after saving the reminder configuration.

These output screens demonstrate that the system is functioning correctly and providing the intended features.

6.2 Performance Evaluation

The system was evaluated based on responsiveness, accuracy, and usability.

- **Response Time:**

The system responds quickly to user requests such as loading event records, applying filters, and displaying time and scheduling summaries.

- **Data Accuracy:**

All time and scheduling data retrieved from the database is displayed correctly and consistently across different modules.

- **Usability:**

The user interface is simple and easy to navigate, allowing users to manage their time managements without confusion.

- **Efficiency:**

The use of JDBC, optimized SQL queries, and efficient API calls ensures efficient data handling and retrieval.

Overall, the system performs well for individual and small-business usage and provides a smooth user experience for managing and analyzing time and scheduling data.

6.3 System Testing

The system was tested using different scenarios to ensure reliability and correctness.

- **Functional Testing:**

All features such as registration, login, event management, event filtering, summary display, and event reminders and alerts were tested and found to be working as expected.

- **Validation Testing:**

Input fields were tested for correct validation, ensuring that invalid or empty data is not accepted.

- **Authentication Testing:**

Login and session management were tested to ensure secure access to the system.

- **Database Testing:**

Data insertion, retrieval, and deletion operations were verified to ensure proper interaction with the database.

- **Error Handling:**

The system handles errors gracefully and displays appropriate messages to the user.

The testing results confirm that the system is stable, reliable, and ready for deployment.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Online Clock and Calendar Web App system was successfully designed and developed to simplify the process of recording, categorizing, and managing date, time, and event information. The application provides a user-friendly interface that allows users to add event and event entries, filter records by category or date, view detailed time and scheduling summaries, and set event reminders and alerts.

The system effectively integrates frontend and backend technologies using Java Servlets, JSP, and MySQL. It follows a structured MVC architecture that ensures clear separation of components, making the application easy to understand and maintain. Key features such as user authentication, event management, reminder alert configuration, and database interaction have been implemented successfully.

The project demonstrates the practical application of web development concepts and provides a reliable platform for managing and analyzing personal time and scheduling data. Overall, the system meets its objectives and performs efficiently for individual and small-business usage.

7.2 Future Enhancements

Although the system meets the current requirements, it can be further improved by adding more advanced features in the future:

- **Advanced Event Filters:**
Adding advanced filters such as custom date ranges, sub-categories, and multi-currency support for more precise time and scheduling analysis
- **Interactive Event Charts:**
Displaying event trends, event-to-event ratios, and category-wise breakdowns on interactive charts for better time and scheduling visualization
- **AI-Based Spending Recommender:**
Providing AI-powered spending recommendations and saving suggestions based on historical event patterns
- **Mobile Application:**
Developing a mobile version of the Online Clock and Calendar Web App for improved accessibility and real-time reminder push notifications
- **Admin Panel:**
Providing an admin interface to manage users, event categories, and system-wide reminder configurations
- **Email Notifications:**
Sending automatic event reminders and alerts via email when monthly spending exceeds the configured threshold
- **Improved Security:**
Enhancing security with encryption, stronger authentication mechanisms, and secure API key management

These enhancements can make the system more scalable, interactive, and suitable for real-world deployment across a wider user base.

REFERENCES

- [1] Oracle Corporation, *Java Servlet Technology Documentation*.
Available: <https://docs.oracle.com/javaee/>
- [2] Oracle Corporation, *JSP (Java Server Pages) Documentation*.
Available: <https://docs.oracle.com/javaee/>
- [3] Oracle Corporation, *JDBC API Documentation*.
Available: <https://docs.oracle.com/javase/>
- [4] Apache Software Foundation, *Apache Tomcat Documentation*.
Available: <https://tomcat.apache.org/>
- [5] MySQL Documentation Team, *MySQL Reference Manual*.
Available: <https://dev.mysql.com/doc/>
- [6] MDN Web Docs, *HTML, CSS, and Web Development Guides*.
Available: <https://developer.mozilla.org/>
- [7] W3Schools, *Web Development Tutorials*.
Available: <https://www.w3schools.com/>
- [8] GeeksforGeeks, *Java and Web Development Articles*.
Available: <https://www.geeksforgeeks.org/>
- [9] K. Sierra and B. Bates, *Head First Servlets and JSP*, O'Reilly Media, 2nd Edition, 2008.
- [10] E. Balagurusamy, *Programming with Java*, McGraw Hill Education, 6th Edition.

APPENDICES

Appendix-1: Sample Database Tables

Users Table

Field Name	Data Type	Description
user_id	INT (PK)	Unique user ID
name	VARCHAR	User name
email	VARCHAR	User email
password	VARCHAR	User password

Transactions Table

Field Name	Data Type	Description
event_id	INT (PK)	Unique location ID
title	VARCHAR	Transaction type (event/event)
price	DOUBLE	Event category
location	VARCHAR	Transaction amount
description	TEXT	Description
image	VARCHAR	Transaction date
user_id	INT (FK)	Reference to user

Reminder Alerts Table

Field Name	Data Type	Description
alert_id	INT (PK)	Unique alert ID
event_id	INT (FK)	Related user
name	VARCHAR	Transaction description
message	TEXT	Reminder limit amount

Appendix-2: Sample SQL Queries

-- Insert User

```
INSERT INTO users (name, email, password)
VALUES ('John Doe', 'john@example.com', '123456');
```

-- Insert Saved Location

```
INSERT INTO events (user_id, type, category, amount, description, trans_date)
VALUES (1, 'Event', 'Food', 500.00, 'Lunch at canteen', '2024-01-15');
```

-- Search Saved Location

```
SELECT * FROM events WHERE
category = 'Food' AND trans_date
BETWEEN '2024-01-01' AND '2024-
01-31';
```

-- Delete Transaction

```
DELETE FROM events
WHERE event_id = 1;
```

Appendix-4: Key Servlet Example

@WebServlet("/index")

```
public class EventServlet extends HttpServlet {
```

```
    protected void doGet(HttpServletRequest request,
                          HttpServletResponse response)
        throws ServletException, IOException {
```

```
        List<Transaction> list = new ArrayList<>();
        Connection con = DBConnection.getConnection();
```

```
        try {
            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery(
                "SELECT * FROM events WHERE user_id = ?");
```

```
            while (rs.next()) {
                Property p = new Property();
                p.setTitle(rs.getString("title"));
                p.setPrice(rs.getDouble("price"));
                list.add(p);
            }
```

```
        } catch (Exception e) {
            e.printStackTrace();
        }
```

```
request.setAttribute("propertyList", list);
request.getRequestDispatcher("index.jsp")
    .forward(request, response);
}
```

PSEUDOCODE

The following pseudocode represents the core functionalities of the online clock and calendar web app system.

1. User Registration

```
BEGIN
  INPUT name, email, password

  CONNECT to database

  CHECK if email already exists
  IF exists THEN
    DISPLAY "Email already registered"
  ELSE
    INSERT user details into users table
    DISPLAY "Registration successful"
  ENDIF

  CLOSE database connection
END
```

2. User Login

```
BEGIN
  INPUT email, password

  CONNECT to database

  FETCH user where email and password match

  IF user exists THEN
    CREATE session
    STORE user details in session
    REDIRECT to homepage
  ELSE
    DISPLAY "Invalid
credentials"
  ENDIF

  CLOSE database connection
END
```

3. Add Transaction

```
BEGIN
  CHECK if user is logged in

  INPUT city_name, latitude, longitude CONNECT to
  database

  INSERT location details into saved_locations
    table with user_id from session
  DISPLAY "Property added successfully"
  CLOSE database connection
END
```

4. View Event Dashboard

```
BEGIN
  CONNECT to database

  FETCH all events for logged-in user

  CALCULATE total event, total events, and
  balance STORE results in dashboardData
  object

  SEND data to JSP page

  DISPLAY clock and calendar
  dashboard to user

  CLOSE database connection
END
```

5. Search Transactions by Category

```
BEGIN
  INPUT category, date_from,
  date_to CONNECT to database

  PARSE API response:
    SELECT * FROM events WHERE
    category LIKE '%keyword%' AND
    trans_date BETWEEN date_from
    AND date_to
```

STORE results in eventList object

DISPLAY filtered events to user

CLOSE database connection
END

6. View Transaction Details

BEGIN
GET event_id from request

CONNECT to database

FETCH full event details including amount,

category, date, description DISPLAY

complete event record to user

CLOSE database connection
END

7. Set Reminder Alert

BEGIN
INPUT alert_type, threshold, user_id

CONNECT to database

INSERT reminder alert preferences into reminder_alerts table
DISPLAY "Inquiry sent successfully"
CLOSE database connection
END

8. Delete Transaction

BEGIN
GET event_id CHECK if

user is owner

CONNECT to database

DELETE related event reminders and alerts if

applicable DELETE event from events table

DISPLAY "Transaction deleted successfully"

```
CLOSE database connection  
END
```



CERTIFICATE OF COMPLETION

JESHWANTH P

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A handwritten signature in black ink.

Krishna Kumar
CEO, Simplilearn



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